Search Smooths Discontinuities in Discrete/Continuous Problems

Christopher D. Carroll Fedor Iskhakov Shujaat Khan Adrian Monninger

JHU

Stony Brook

IMF

Texas A&M

September 15, 2026

Discontinuities break EGM in discrete–continuous problems

Problem:

- EGM is efficient when the choice is continuous and the value function is concave
- With discrete *and* continuous choices, the Euler equation is only necessary
- Kinks in V and jumps in the policy create non-monotonic endogenous grids

$\rightarrow$ suboptimal EGM points; kinks propagate under backward induction

Discontinuities break EGM in discrete–continuous problems

Problem:

- EGM is efficient when the choice is continuous and the value function is concave
- With discrete *and* continuous choices, the Euler equation is only necessary
- Kinks in V and jumps in the policy create non-monotonic endogenous grids

$\rightarrow$ suboptimal EGM points; kinks propagate under backward induction

Standard fix: Extreme Value Type I taste shocks

- McFadden (discrete choice); Iskhakov et al. (2017) for consumption–saving DC-EGM
- Smooths V and marginal utilities, but randomizes the discrete outcome

Discontinuities break EGM in discrete–continuous problems

Problem:

- EGM is efficient when the choice is continuous and the value function is concave
- With discrete *and* continuous choices, the Euler equation is only necessary
- Kinks in V and jumps in the policy create non-monotonic endogenous grids

$\rightarrow$ suboptimal EGM points; kinks propagate under backward induction

Standard fix: Extreme Value Type I taste shocks

- McFadden (discrete choice); Iskhakov et al. (2017) for consumption–saving DC-EGM
- Smooths V and marginal utilities, but randomizes the discrete outcome

This paper: directed search + price dispersion

- Search effort governs the probability of switching the discrete state
- Price (or account) draws rationalize when the switch is exercised
- Smoothness is part of the agent's economic problem—not only a numerical trick

Why search and prices instead of taste shocks?

1. **Local to the threshold**
   Search affects agents near S,s (or work–retire) bands; taste shocks put mass on dominated branches far from indifference

2. **Microfounded and measurable**
   Search intensity $\varepsilon^* \in [0, 1]$ maps to expected search duration — observable in microdata
   Taste-shock scale σ_ε is hard to interpret (match far-from-threshold adjusters or iMPCs)

3. **Compatible with permanent-income normalization**
   Search costs can scale with permanent income; additive EV shocks typically cannot $\Rightarrow$ extra state

Why search and prices instead of taste shocks?

1. **Local to the threshold**
   Search affects agents near S,s (or work–retire) bands; taste shocks put mass on dominated branches far from indifference

2. **Microfounded and measurable**
   Search intensity $\varepsilon^* \in [0, 1]$ maps to expected search duration — observable in microdata
   Taste-shock scale σ_ε is hard to interpret (match far-from-threshold adjusters or iMPCs)

3. **Compatible with permanent-income normalization**
   Search costs can scale with permanent income; additive EV shocks typically cannot $\Rightarrow$ extra state

Complementarity: search alone is one-sided; price dispersion (or small taste shocks) delivers uniform smoothness around discontinuities

Retirement model: environment

- Utility: $u(c_t, d_t, \varepsilon_t) = \log(c_t) - \delta d_t - \theta(\varepsilon_t)$
- Search cost: $\theta(\varepsilon_t) = \sigma_s\big(\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\big)$
- Discrete state: work ($d_t = 1$) or retire ($d_t = 0$), retirement absorbing
- Budget: $M_t = RA_{t-1} + y_t d_{t-1}$, with $y_t = y\eta_t$, η_t mean-one lognormal
- Option to retire arrives with probability $P_1(\varepsilon_{t-1}) = (1-\varepsilon_{t-1})p + \varepsilon_{t-1}$

Retirement model: environment

- Utility: $u(c_t, d_t, \varepsilon_t) = \log(c_t) - \delta d_t - \theta(\varepsilon_t)$
- Search cost: $\theta(\varepsilon_t) = \sigma_s\big(\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\big)$
- Discrete state: work ($d_t = 1$) or retire ($d_t = 0$), retirement absorbing
- Budget: $M_t = RA_{t-1} + y_t d_{t-1}$, with $y_t = y\eta_t$, η_t mean-one lognormal
- Option to retire arrives with probability $P_1(\varepsilon_{t-1}) = (1-\varepsilon_{t-1})p + \varepsilon_{t-1}$

Retiree (absorbing):

$$V_t^{\mathrm{Ret}}(M_t) = \max_{0 \le c_t \le M_t} \left\{ \log(c_t) + \beta V_{t+1}^{\mathrm{Ret}}(RA_t) \right\}$$

Retirement model: worker's problem

Must-work branch $\mathcal{D}_t = \{W\}$:

$$V_t(M_t, \{W\}, \xi_t) = \xi_t \max_{\substack{0 \le c_t \le M_t \\ 0 \le \varepsilon_t \le 1}} \left\{ u(c_t, 1, \varepsilon_t) + \beta P_1(\varepsilon_t)\, \mathcal{V}_t^1(A_t) + \beta P_2(\varepsilon_t)\, \mathcal{V}_t^2(A_t) \right\}$$

Retirement model: worker's problem

Must-work branch $\mathcal{D}_t = \{W\}$:

$$V_t(M_t, \{W\}, \xi_t) = \xi_t \max_{\substack{0 \le c_t \le M_t \\ 0 \le \varepsilon_t \le 1}} \left\{ u(c_t, 1, \varepsilon_t) + \beta P_1(\varepsilon_t)\, \mathcal{V}_t^1(A_t) + \beta P_2(\varepsilon_t)\, \mathcal{V}_t^2(A_t) \right\}$$

With option $\mathcal{D}_t = \{R, W\}$:

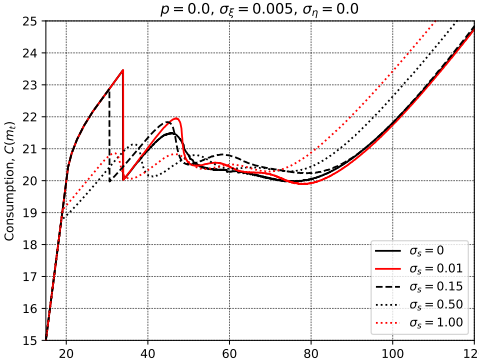
$$V_t(M_t, \{R, W\}, \xi_t) = \max \left\{ V_t^{\mathrm{Ret}}(M_t);\ V_t(M_t, \{W\}, \xi_t) \right\}$$

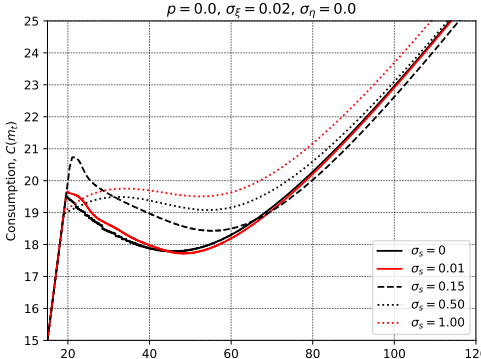
Optimal effort (after consumption):

$$\varepsilon_t^* = 1 - \exp\left(-\frac{\beta(1-p)}{\sigma_s} \left(\mathcal{V}_t^1(A_t) - \mathcal{V}_t^2(A_t) \right) \right)$$

Retirement: multiplicative log-normal taste shocks

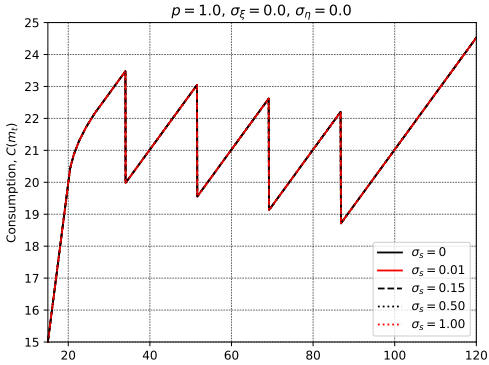
- $\xi_t \sim \mathcal{N}(-\sigma_\xi^2/2, \sigma_\xi^2)$ scales the value of working
- Unlike EV Type I (Iskhakov et al.), these shocks are discretized and feed into search effort
- Alone: expand the region where agents search around discontinuities

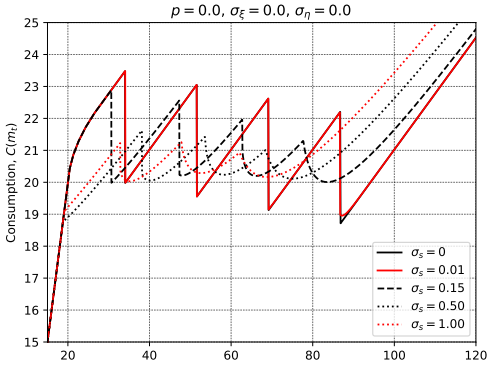




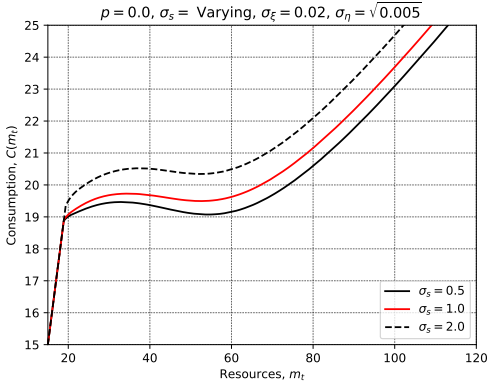
Retirement: directed search smooths (asymmetrically)

- $p = 1$: option arrives for free $\Rightarrow \varepsilon^* = 0$, no smoothing from σ_s
- $p = 0$: must search to retire $\Rightarrow$ larger σ_s smooths to the *right* of kinks
- Low-M workers optimally set $\varepsilon^* = 0$ (plan to keep working)

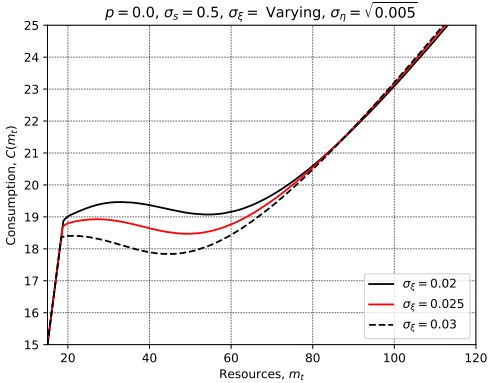




Retirement: search + taste shocks ⇒ uniform smoothness



Raising σ_s shifts c **up**



Raising σ_ξ shifts c **down**

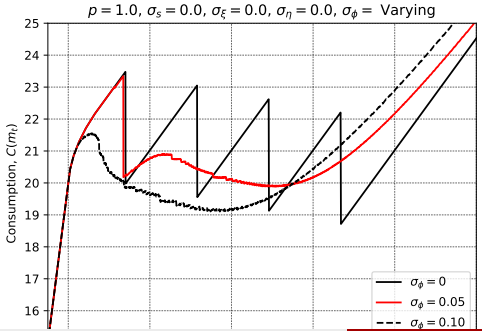
- Search / lower p: fewer successful switches ⇒ more expected work years ⇒ higher c

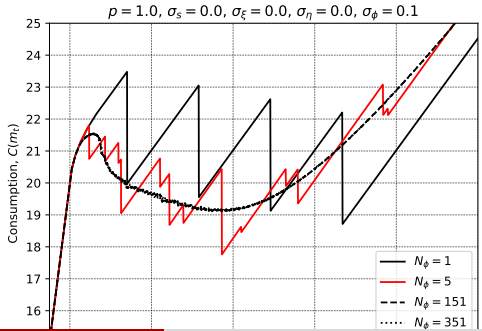
Retirement: stochastic account realizations

After receiving the option, draw rejectable retirement wealth factor ϕ_t:

$$\log \phi_t \sim \mathcal{N}\big(-\frac{\sigma_\phi^2}{2}, \sigma_\phi^2\big), \quad \tilde{M}_t = \phi_t M_t, \quad V_t(M_t, \phi_t) = \max\big\{V_t^{\text{Ret}}(\phi_t M_t),\ V_t^W(M_t)\big\}$$

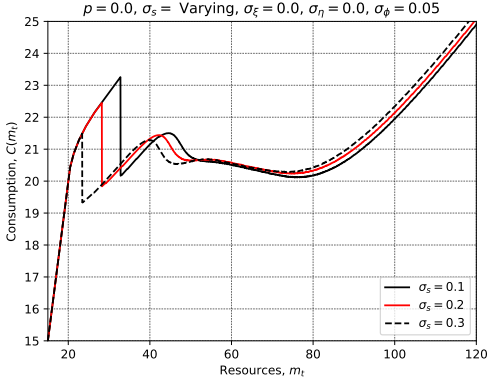
Ex-ante value $\bar{V}_t(M_t) = \mathbb{E}_{\phi_t}[V_t(M_t, \phi_t)]$ replaces a single threshold M^* by a family $M^*(\phi)$.

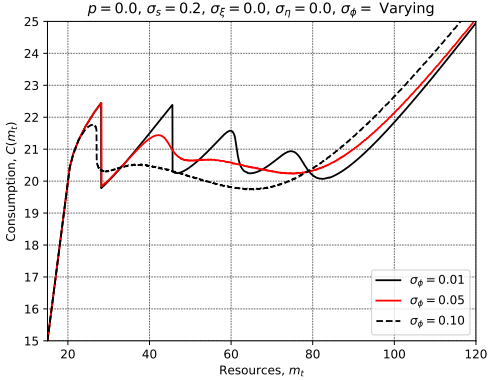




Retirement: search and account dispersion together

- Search governs *whether* a draw arrives; ϕ_t is the *content* of the draw
- Complements: option value of ϕ raises $\mathcal{V}^1 - \mathcal{V}^2 \Rightarrow$ more search





Durable model: keep vs adjust

$$u(c_t, d_t) = \frac{(c_t^{\alpha} d_t^{1-\alpha})^{1-\rho}}{1-\rho}, \qquad \ell_t = m_t + (1-\tau) n_t$$

Hard maximum:

$$v_t(m_t, n_t) = \max\left\{ v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t) \right\}$$

Durable model: keep vs adjust

$$u(c_t, d_t) = \frac{(c_t^{\alpha} d_t^{1-\alpha})^{1-\rho}}{1-\rho}, \qquad \ell_t = m_t + (1-\tau)n_t$$

Hard maximum:

$$v_t(m_t, n_t) = \max\left\{v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t)\right\}$$

Keeper:

$$v_t^{\text{keep}} = \max_{c_t} u(c_t, n_t) + \beta \mathbb{E}_t[v_{t+1}]$$

$$a_t = m_t - c_t \geq 0, \quad d_t = n_t$$

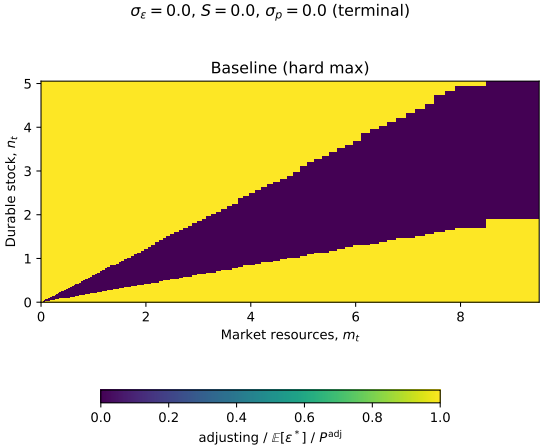
Adjuster:

$$v_t^{\text{adj}} = \max_{c_t, d_t} u(c_t, d_t) + \beta \mathbb{E}_t[v_{t+1}]$$

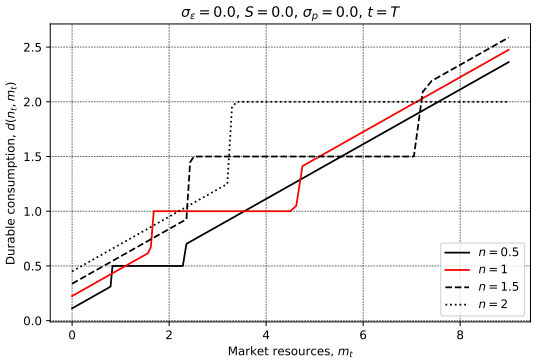
$$a_t = \ell_t - c_t - d_t \geq 0$$

Laws of motion: $m_{t+1} = \mathcal{R} a_t + \theta_{t+1},\ n_{t+1} = (1-\delta) d_t$

Baseline hard max: S,s bands and discontinuous policies



Terminal keep–adjust regions



Combined durable policy

Durable model: EV Type I taste shocks

$$v_t = \sigma_\varepsilon \log\left(\exp\left(v_t^{\text{keep}}/\sigma_\varepsilon\right) + \exp\left(v_t^{\text{adj}}/\sigma_\varepsilon\right)\right)$$

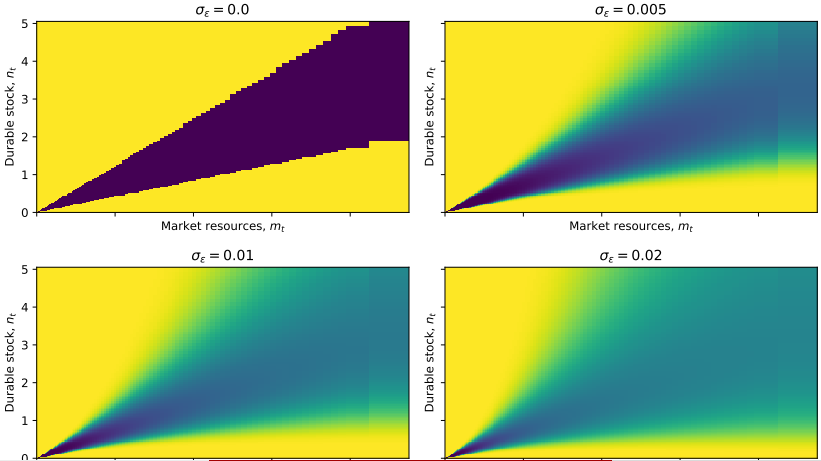
Logit probabilities and mixed policies:

$$P_t^{\text{adj}} = \frac{\exp(v_t^{\text{adj}}/\sigma_\varepsilon)}{\exp(v_t^{\text{keep}}/\sigma_\varepsilon) + \exp(v_t^{\text{adj}}/\sigma_\varepsilon)},$$
$$c_t = P_t^{\text{keep}}\, c_t^{\text{keep}} + P_t^{\text{adj}}\, c_t^{\text{adj}}, \quad d_t = P_t^{\text{keep}}\, n_t + P_t^{\text{adj}}\, d_t^{\text{adj}}$$

- Closed form (no taste quadrature); continuous FOCs unchanged
- $\sigma_\varepsilon \to 0$ recovers hard max

Taste shocks: terminal S,s bands

$\sigma_\varepsilon \in \{0, 0.05, 0.1, 0.2\}$, $S = 0$, $\sigma_p = 0$ (terminal)



Durable model: directed search

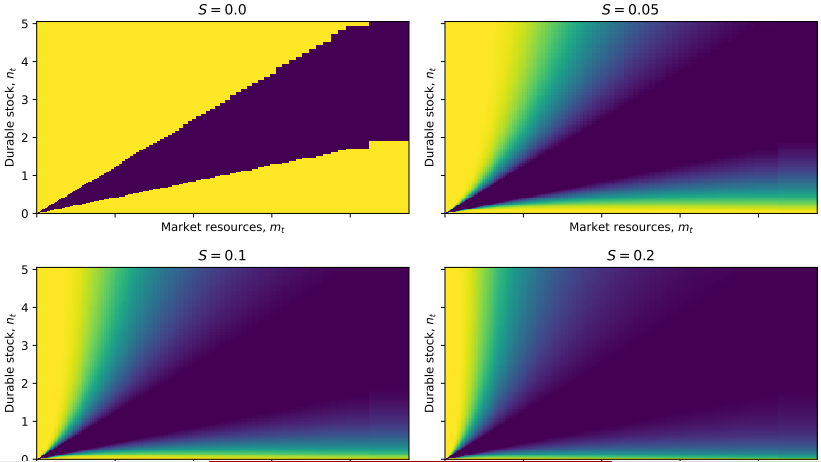
$$\sigma(\varepsilon_t) = S\big[\varepsilon_t + (1-\varepsilon_t)\log(1-\varepsilon_t)\big]$$

$$v_t(m_t, n_t) = \max_{\varepsilon_t \in [0,1]} \Big[(1-\varepsilon_t)\, v_t^{\text{keep}} + \varepsilon_t\, v_t^{\text{adj}} - \sigma(\varepsilon_t)\Big]$$

- ε_t: probability of locating a suitable durable this period
- $S \to 0$: hard max; $S \to \infty$: never adjusts
- Effort chosen before the discrete outcome — economically local near the band

Search: terminal S,s bands

$\sigma_\varepsilon = 0$, $S \in \{0, 0.05, 0.1, 0.2\}$, $\sigma_p = 0$ (terminal)



Durable model: stochastic purchase prices

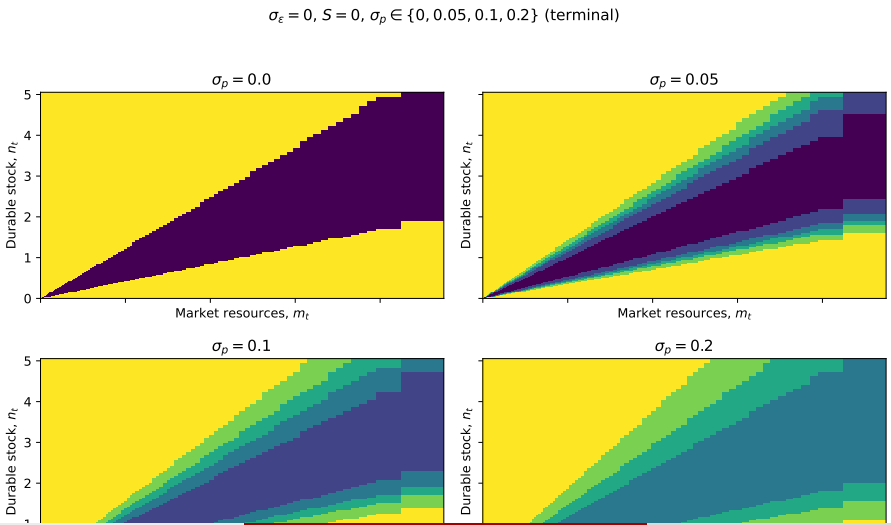
$$\log p_t \sim \mathcal{N}\big(-\tfrac{\sigma_p^2}{2}, \sigma_p^2\big), \quad a_t = \ell_t - c_t - p_t\, d_t \geq 0$$

Price-contingent hard max and ex-ante value:

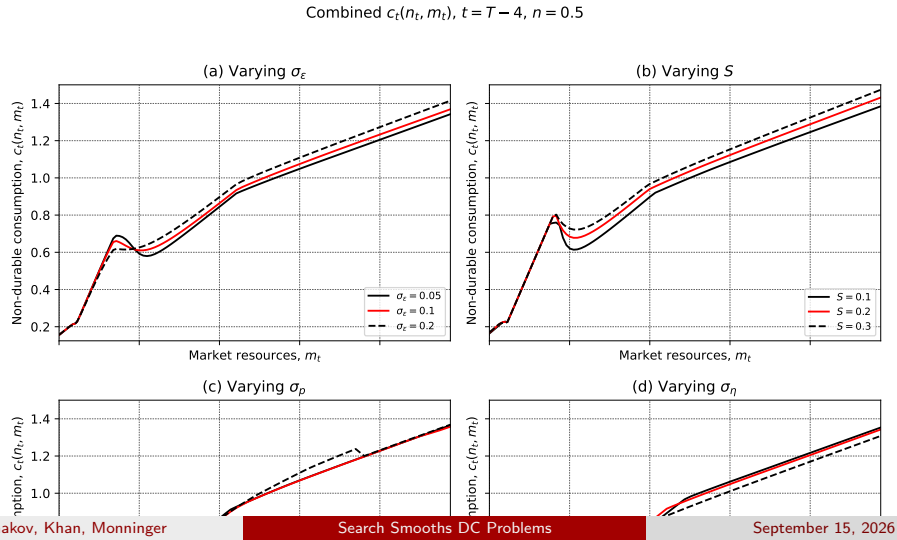
$$v_t(m_t, n_t, p_t) = \max\left\{v_t^{\text{keep}}(m_t, n_t),\ v_t^{\text{adj}}(\ell_t, p_t)\right\}$$
$$\bar{v}_t(m_t, n_t) = \mathbb{E}_{p_t}\big[v_t(m_t, n_t, p_t)\big]$$

- Only *new* durable purchases face p_t; liquidation still at rate $1 - \tau$
- Changes adjuster's continuous FOC in d_t (unlike EV shocks)
- Low p_t accepted, high p_t rejected — softens the extensive margin

Price dispersion: terminal adjustment regions



Combined nondurable consumption: all channels move together

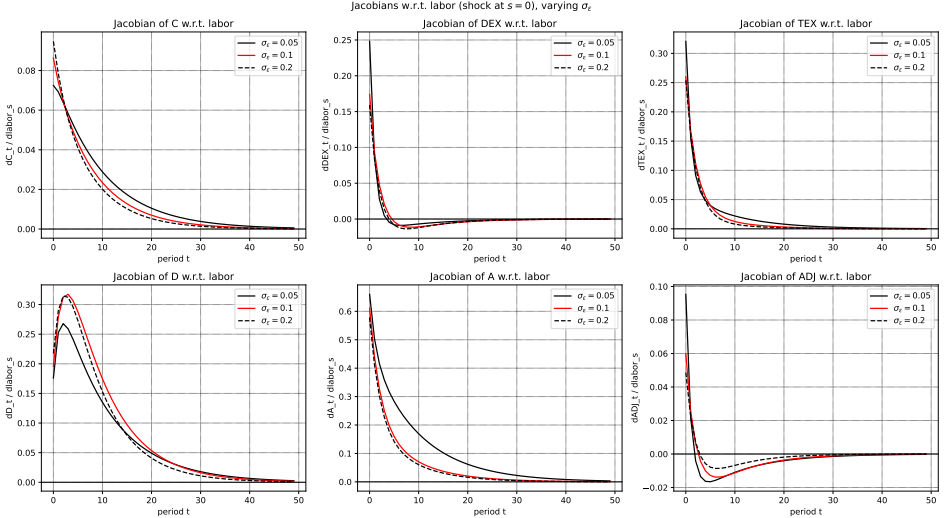


Infinite horizon: steady-state aggregates

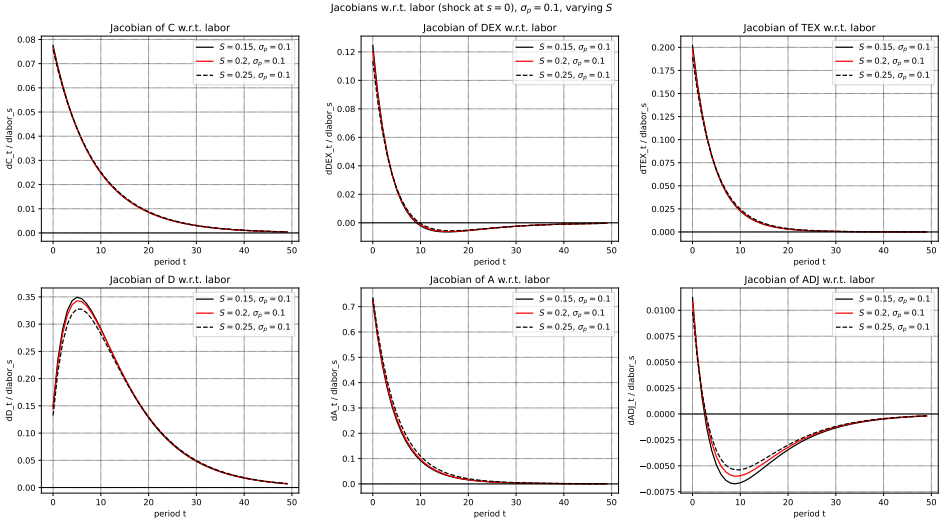
		Steady state					
		C	D	DEX	TEX	A	ADJ (%)
σ_ε	0.05	0.82	3.38	0.17	0.99	1.74	9.8
	0.1	0.80	3.31	0.17	0.97	1.25	13.5
	0.2	0.79	2.97	0.15	0.94	0.66	25.0
σ_p ($S = 0.2$)	0.08	0.83	4.35	0.22	1.05	1.60	9.8
	0.1	0.83	4.87	0.24	1.07	1.35	11.6
	0.12	0.83	5.64	0.28	1.11	1.18	13.8
S ($\sigma_p = 0.1$)	0.15	0.83	4.98	0.25	1.08	1.28	12.4
	0.2	0.83	4.87	0.24	1.07	1.35	11.6
	0.25	0.83	4.83	0.24	1.07	1.47	10.8

- Taste vs search+price: higher smoothing $\Rightarrow$ lower C/DEX
- Adjuster share: $\uparrow$ with taste or σ_p; $\downarrow$ with S (given σ_p)

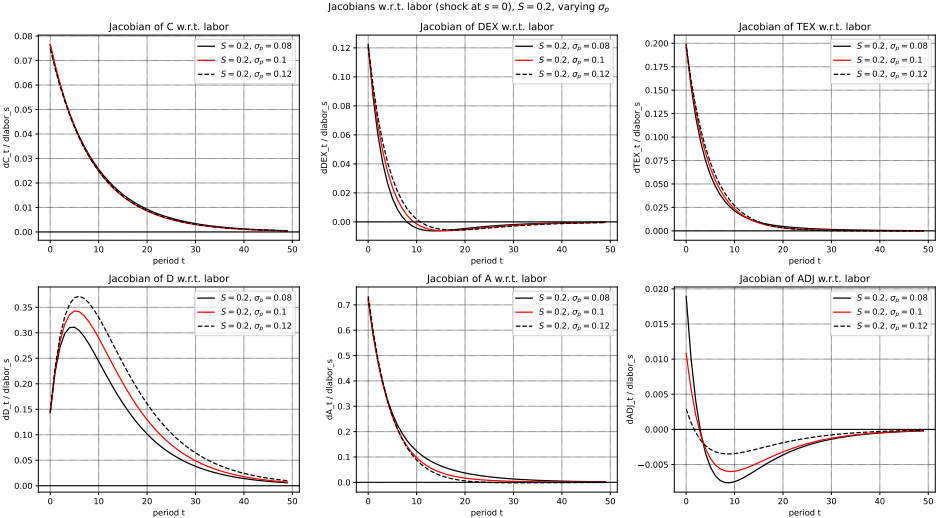
Jacobians w.r.t. labor: taste shocks



Jacobians w.r.t. labor: varying search intensity



Jacobians w.r.t. labor: varying price dispersion



Takeaways

1. Discrete–continuous EGM produces suboptimal points from policy discontinuities
2. Directed search + price (account) dispersion smooth those discontinuities while keeping an economic interpretation
3. Search is preferable to taste shocks when locality, microfoundation, and PIH normalization matter
4. In durables, all smoothers shift combined c the same way; adjuster shares move differently (taste/price vs search)
5. With enough smoothness, upper-envelope cleanup can become unnecessary

Thank you

Repo: `https://github.com/AMonninger/DCEGM_Search`